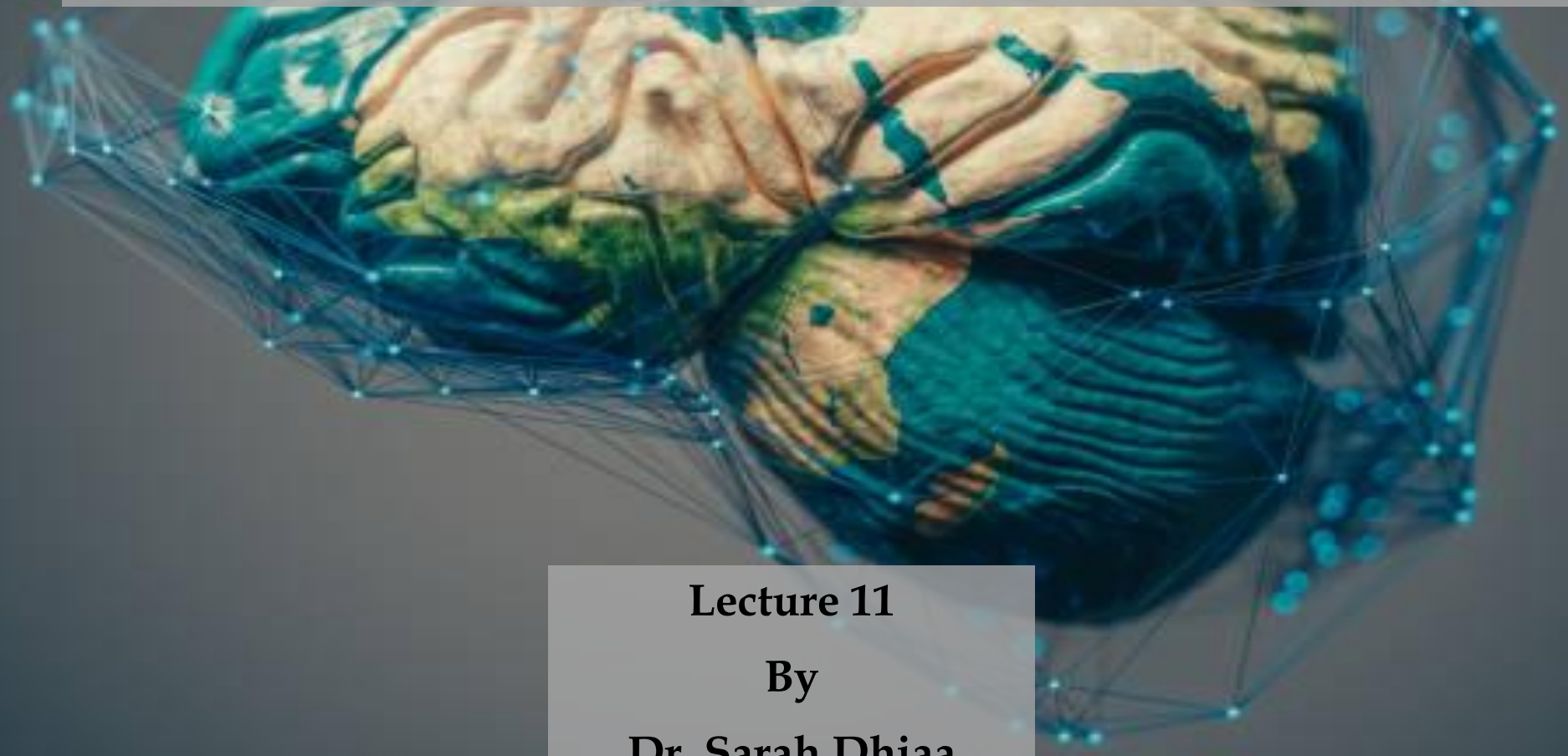
The Extra Pyramidal System



Lecture 11

By

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Objectives:

THE EXTRA PYRAMIDAL SYSTEM

- The Upper And Lower Motor Neurons

THE BASAL GANGLIA

- Neural Connections Of The Basal Ganglia
- Neurotransmitters In The Basal Ganglia
- Functions Of The Basal Ganglia
- Lesions In The Basal Ganglia

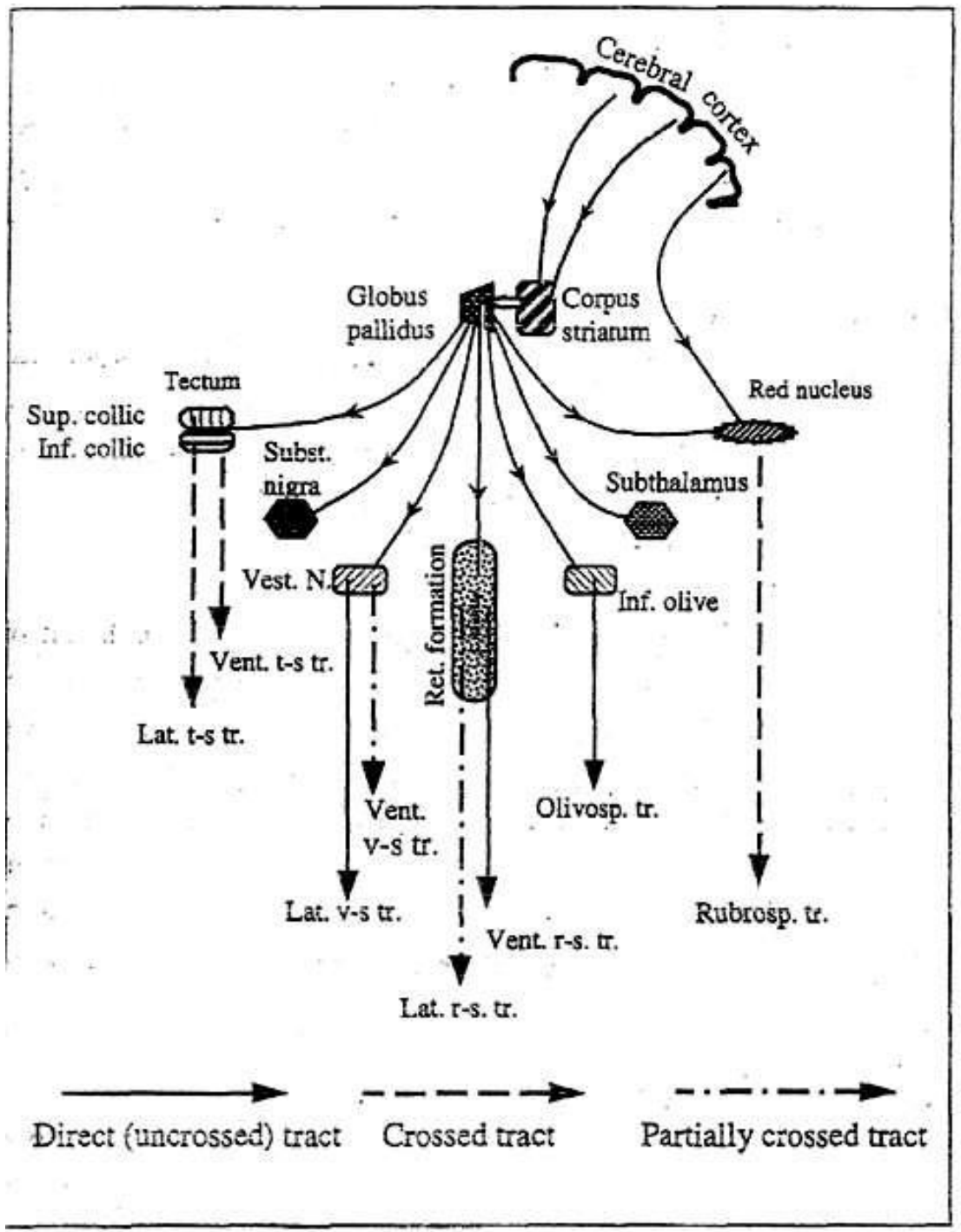
THE CEREBELLUM

- Divisions Of The Cerebellum
- Connections To The Cerebellum
- Functions Of The Cerebellum
- The Cerebellar Syndrome

THE EXTRA PYRAMIDAL SYSTEM

- The extrapyramidal system includes all parts of the nervous system other than the pyramidal tracts that contribute to the control of skeletal muscle activity.
- These parts include the basal ganglia, the red nucleus in the midbrain , the reticular formation of the brainstem, the vestibular nuclei of the pons and medulla , and the tectum of the midbrain and the inferior olive of the medulla.
- All these parts send signals down to the spinal motor nuclei in the descending extrapyramidal tracts.

- **THE BASAL GANGLIA**
- It receives projection fibers from the motor cortex to corpus striatum and then to globus pallidus then to several nuclei in the brainstem


Cerebral cortex
Globus pallidus
Corpus striatum
Tectum
Sup. collic
Inf. collic
Subst. nigra
Red nucleus
Subthalamus
Vest. N.
Ret. formation
Inf. olive
Vent. t-s tr.
Lat. t-s tr.
Vent. v-s tr.
Olivosp. tr.
Lat. v-s tr.
Vent. r-s. tr.
Rubrosp. tr.
Lat. r-s. tr.
Direct (uncrossed) tract
Crossed tract
Partially crossed tract

THE RED NUCLEUS

- It is located in the midbrain and receives projection fibers from the motor cortex **(the corticorubral tract)** and collaterals from the corticospinal tract as it passes through the midbrain. It also receives projection fibers from the globus pallidus of the basal ganglia.
- All these fibers synapse in the lower part of the red nucleus.
- **Rubrospinal tract** originates from these neurons, cross to the opposite side and descend in the spinal cord very closely anterior to the lateral corticospinal tract.

THE TECTUM OF THE MIDBRAIN

- It receives projection fibers from the globus pallidus of the basal ganglia, and gives origin to two descending extrapyramidal tracts:
- **The lateral tectospinal tract**: Originates from the superior colliculus (the center of visual reflexes), crosses to the opposite side and terminates in the cervical segments of the spinal cord.
- It is concerned with directing the eye and turning the head towards a light source (visuospinal reflexes).

* **The ventral tectospinal tract:** Originates from the inferior colliculus (the center of auditory reflexes), crosses to the opposite side and terminates in the cervical segments of the spinal cord.

- It is concerned with turning the head to direct the ears towards a sound source (audiospinal reflexes).

THE RETICULAR FORMATION OF THE BRAINSTEM

- It receives projection fibers from the globus pallidus of the basal ganglia, and gives origin to two descending extrapyramidal tracts:

* **The lateral reticulospinal tract:** Originates from the inhibitory reticular formation of the medulla. Some fibers cross to the opposite side, but most fibers descend in the same side of the spinal cord.

- It inhibits the stretch reflex and skeletal muscle tone.
- **The ventral reticulospinal tract:** Originates from the facilitatory reticular formation of the pons. Fibers descend without crossing to terminate on the same side of the spinal cord.
- It facilitates the stretch reflex and the skeletal muscle tone.

THE VESTIBULAR NUCLEUS OF THE MEDULLA

- It receives projection fibers from the globus pallidus of the basal ganglia, and gives origin to two descending extrapyramidal tracts:

* **The lateral vestibulospinal tract:** Originates from the vestibular nucleus, descends uncrossed to terminate on the alpha and gamma motor neurons (postural adjustment)

* **The ventral vestibulospinal tract:** Originates from the vestibular nucleus, descends on both sides of the spinal cord to terminate on the alpha and gamma motor neurons (head position adjustment)

- The vestibulospinal tracts facilitate the stretch reflex and skeletal muscle tone.

THE INFERIOR OLIVE OF THE MEDULLA

- The inferior olive receives input fibers from **the** motor cortex, **the** globus pallidus of the basal ganglia and **the** spinal cord. It sends output fibers to the cerebellum.
- It projects the fibers of the **olivospinal tract** which descend in the spinal cord to terminate on the ventral horn cells of the same side.
- The olivospinal tract is facilitatory to the stretch reflex and the skeletal muscle tone.
- The inferior olive works in close association with the cerebellum to correct any deviation of the muscle contraction from the preset plan of movement.

THE UPPER AND LOWER MOTOR NEURONS

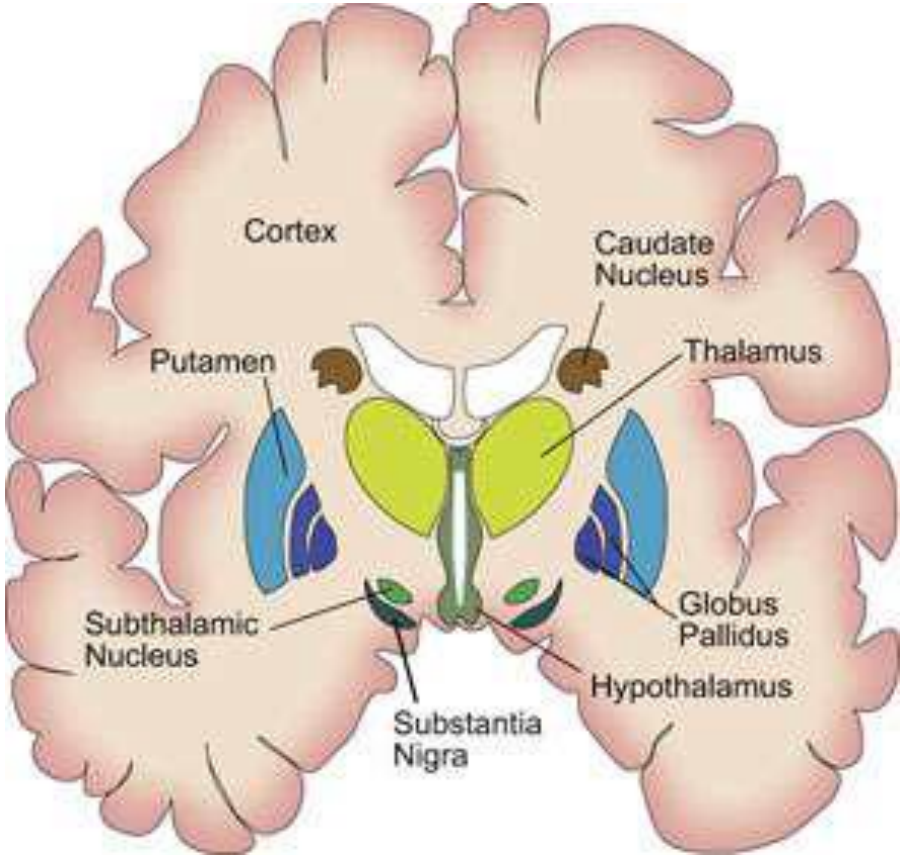
- In the voluntary movement, signals start in the motor neurons of the cerebral cortex and reach the skeletal muscles through two orders of neurons:

1. **Upper motor neurons**: These are the neurons of the pyramidal and extrapyramidal tracts in the CNS. They extend from the cerebral cortex and the extrapyramidal nuclei down to the motor neuron pool of the brainstem and spinal cord.
2. **Lower motor neurons:** These are the neurons of the motor neuron pool and their axons which form the motor nerves to the skeletal muscles.

THE BASAL GANGLIA

- The basal ganglia are sub cortical masses of gray matter found in the cerebral hemispheres lateral to the thalamus that includes:

1. The **caudate** nucleus.
2. The **lentiform** (lenticular) nucleus. This consists of 2 parts:
   a. An outer part called the **putamen**.
   b. An inner part called the **globus pallidus**.
3. The **subthalamic** nucleus.
4. The **substantia nigra** in the midbrain.



Neural Connections Of The Basal Ganglia

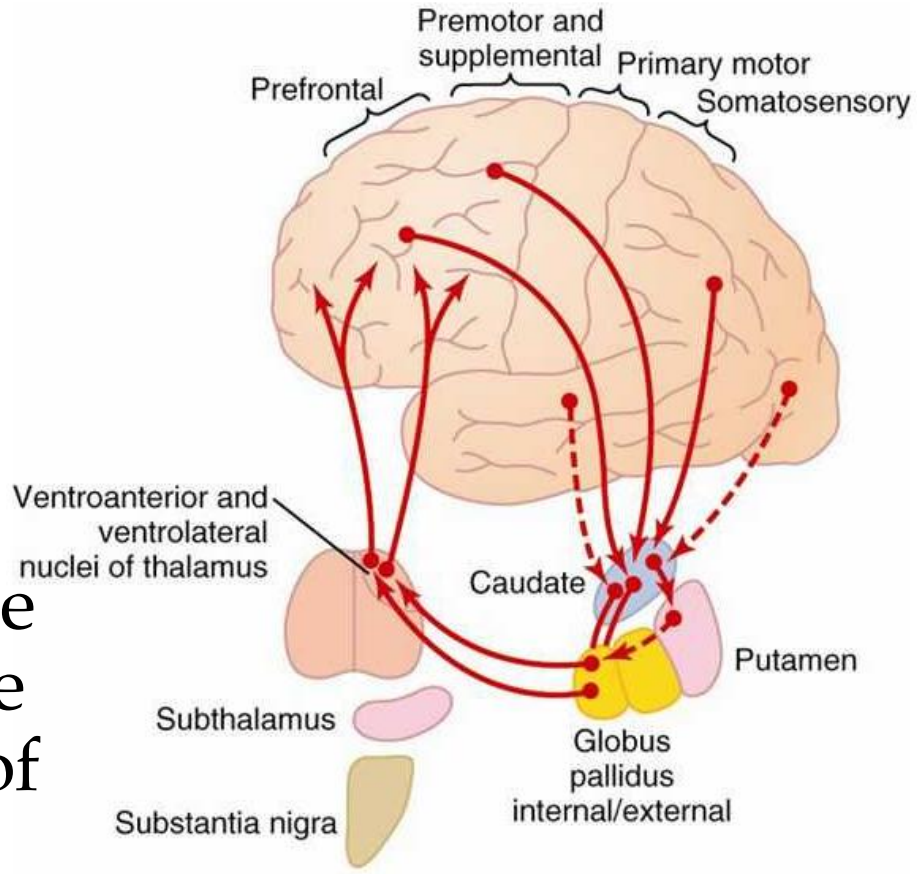
- There are numerous free interconnections between various nuclei of the basal ganglia. This is in addition to the connections with the cerebral cortex, thalamus and brainstem.

A. Connections With The Cerebral Cortex

- through two main circuits, the caudate and putamen circuits.

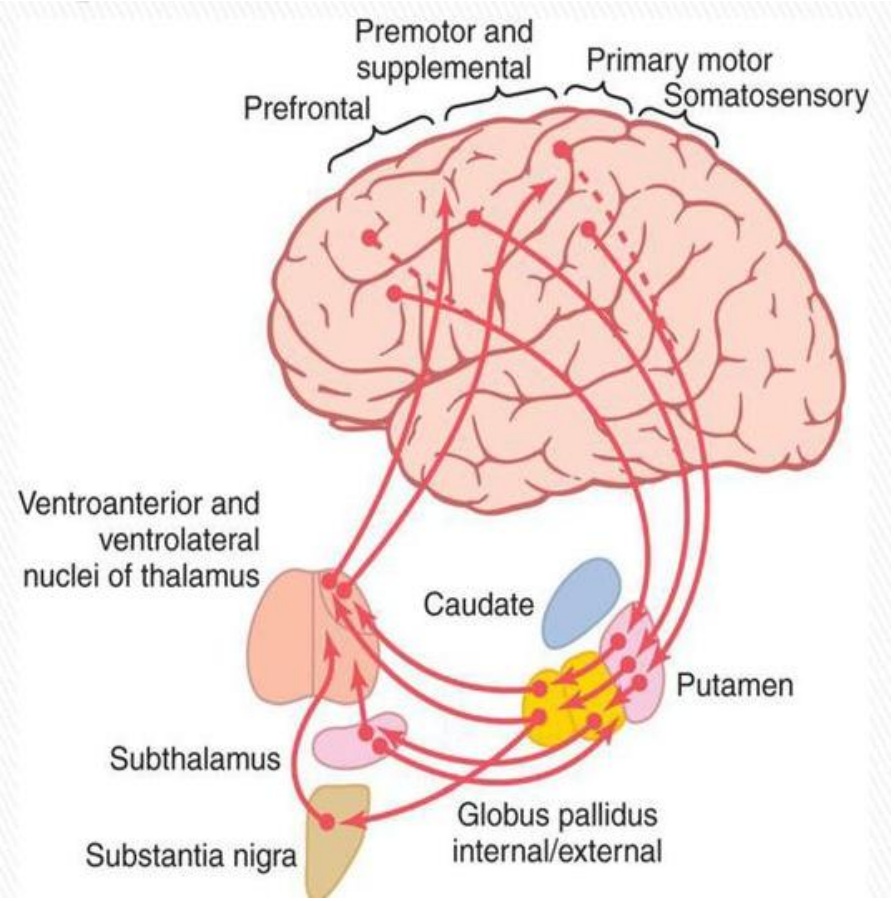
THE CAUDATE CIRCUIT

The caudate circuit starts at the motor association and sensory association areas of the cerebral cortex … to the caudate nucleus … the globus pallidas … ventrolateral (VL) nucleus of the thalamus … premotor and primary motor areas of the cortex.



THE PUTAMEN CIRCUIT

- The putamen circuit starts at the motor association and sensory association areas … to the putamen … the globus pallidus … the ventrolateral (VL) nucleus of the thalamus … premotor and primary motor areas of the cortex.
- **Role** of the putamen circuit in storage of motor programs of familiar motor actions: e.g. Writing one's name or signature, lighting a candle.



Damage of the caudate circuit leads to:

- Inability to organize movements to achieve a complex goal; e.g. one wears the necktie before the shirt.
- Inability to write or draw figures with a fixed scale
- Excluding the contralateral side of the body from any motor plans.

Damage of the putamen circuit leads to:

- **motor apraxia** i.e. loss of ability to carry out familiar movements in the absence of muscle paralysis

B. Connections With The Brainstem

- The basal ganglia are connected with some centers in the brainstem, from which fibers descend in the extrapyramidal tracts down to the spinal cord.
- Fibers project from the globus pallidas to:

1. **The reticular formation**: from which the reticulospinal tracts originate.
2. **The red nucleus:** from which the rubrospinal tract originates.
3. **The vestibular nucleus:** from which the vestibulospinal tracts originate.
4. **The inferior olive:** from which the olivospinal tract originates.

- This is in addition to fibers to the subthalamic, substantia nigra, and tectum.

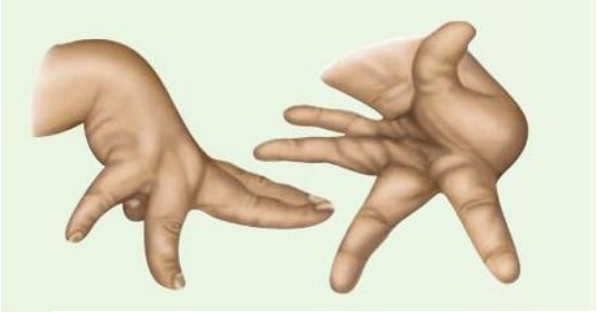
Neurotransmitters In The Basal Ganglia

- Glutamate, acetylcholine and noradrenaline are excitatory transmitters to the basal ganglia.
- dopamine, GABA, serotonin and enkephalin are inhibitory transmitters to the basal ganglia.
- A delicate balance between the excitatory and inhibitory transmitters is essential for proper functioning of the basal ganglia.

Functions Of The Basal Ganglia

- The functions of basal ganglia are purely motor.
- In humans, they play an important role in controlling the muscle tone and voluntary movements.

Manifestations Of Lesions In The Basal Ganglia



- Lesions in the basal ganglia produce two characteristic features which appear on the contralateral side of the lesion:

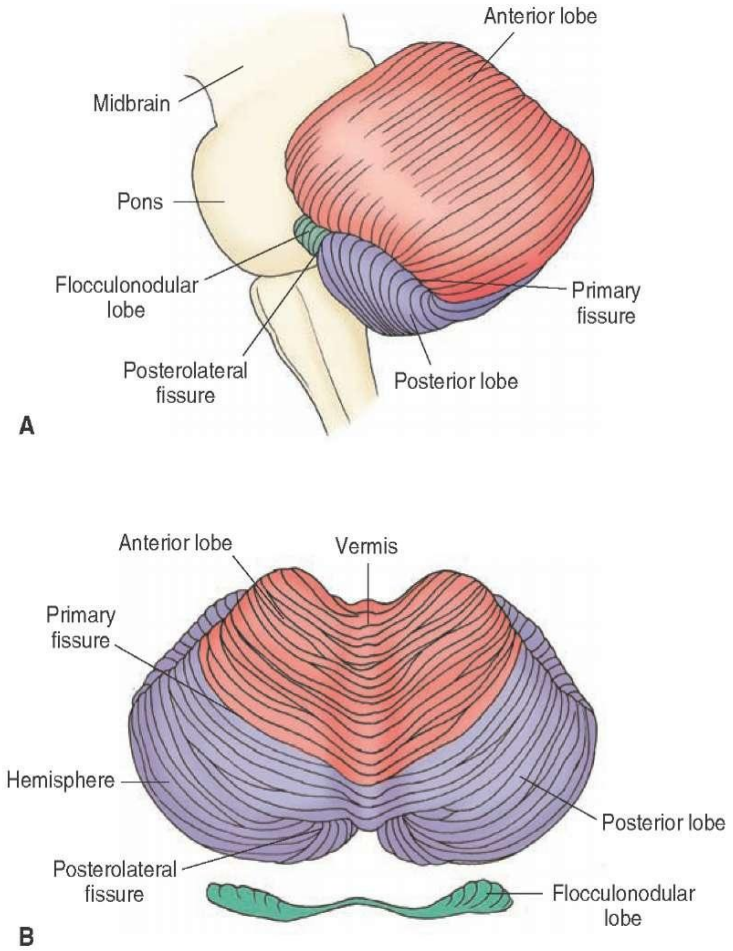
1. **Involuntary movements during rest:** These movements are in the form of chorea (dancing), athetosis (spasmodic not fixed), hemiballismus (jumping) or static tremor (as in Parkinson).
   * The movements disappear during sleep but increase with nervous excitement.
2. **Change in muscle tone:** either hyper or hypotonia.

THE CEREBELLUM

- The cerebellum receives direct or indirect input signals from all the sensory pathways, This sensory information does not reach conscious perception.

ANATOMY

- Cerebellum consists of a midline structure called the **vermis**, two **cerebellar hemispheres** on both sides, and a separate lobe called the **flocculonodular lobe** on the inferior surface.
- Three pairs of **cerebellar peduncles** connect the cerebellum with the brainstem,
  the superior cerebellar peduncle **(SCP)** connects the cerebellum with the midbrain,
  the middle peduncle **(MCP)** with the pons and the inferior peduncle **(ICP)** with the medulla.

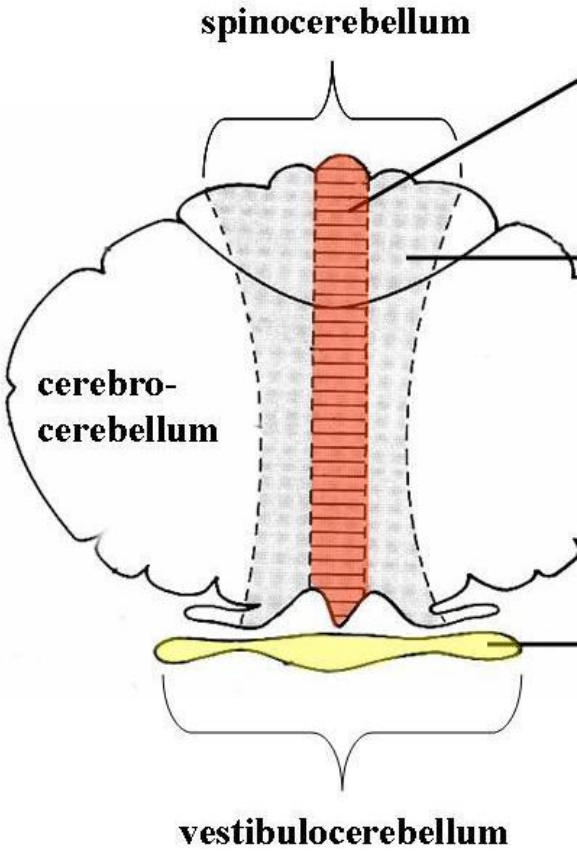


Physiological Divisions Of The Cerebellum

Physiologically, the cerebellum is divided into three functional divisions:

- **Vestibulocerebellum**: It consists of the flocculonodular lobe. It is connected to the vestibular apparatus through the vestibular nuclei in the medulla, It is concerned with **maintenance of equilibrium.**
- **Spinocerebellum**: It consists of the vermis and the medial paravermal parts of the cerebellar hemispheres. It receives signals from the peripheral proprioceptors. It is concerned with the control of **muscle tone and gross movement**.
- **Cerebrocerebellum**: It consists of the lateral parts of the cerebellar hemispheres. It is connected to the cerebral cortex. It is concerned with **the control of skilled voluntary movements** initiated by the cerebral cortex.

* All these three divisions are interconnected by interneurons



Afferent Connections To The Cerebellum

I. From the brain

1) Tecto-cerebellar fibers
2) Cortico ponto cerebellar fibers
3) Olivo cerebellar fibers
4) Vestirulo cerebellar fibers
5) Reticulo cerebellar fibers

II. From the periphery

1) The dorsal spinocerebellar pathway
2) The ventral spinocerebellar pathway

Efferent Connections From The Cerebellum

I. From the **vestibulocerebellum**: fibers from the flocculonodular lobe connected to the vestibular apparatus.

II. From the **spinocerebellum**: fibers from the nuclei in the vermis and paravermal to the reticular formation, motor cortex, and basal ganglia.

III. From the **cerebrocerebellum**: fibers from the nuclei in the cerebellar hemisphere to the thalamus and cerebral cortex.

- The cerebellum is connected to the cerebral cortex by afferent and efferent fibers.
- These form a neuronal circuit that start and end in the cerebral cortex is called the **cortieo ponto cerebello dentate thalamo cortical circuit.**

Functions Of The Cerebellum

The cerebellum is concerned only with subconscious control of motor activity, and its functions include the following:

I. **Control of postural movements and equilibrium** By vestibulocerebellum and some portions of the vermis.

II. **Control of muscle tone and stretch reflex** The cerebrocerebellum and spinocerebellum(mainly) facilitate the muscle tone.

- **The cerebellar stretch reflex (the load reflex)**
  This reflex operates when a contracting muscle meets an unexpected load. The spinocerebellum responds by output signals to the alpha and gamma motor neurons of the spinal cord to increase the muscle tone (the load reflex).

III. Control Of Voluntary Movement

The cerebrocerebellum and the spinocerebellum influence voluntary movement through the following functions:

1. Planning:

- The cerebrocerebellum receives input signals from the premotor cortex and inferior olive before the start of movement to give the cerebellum an "**efference copy**" of the motor plans of the cerebrocerebellum is informed in the **planning of movements.**
- Through connections with the motor cortex the cerebellum "**monitors**" the flow of signals from the motor cortex and corrects any deviation from the planned sequential order.
- **2. Timing of movement:**
- The cerebrocerebellum determines the start and termination of each of the sequential movement which form a complex movement. Without this function decomposition and incoordination of movement.

3. **Joining of movement:**

- The cerebrocerebellum is not concerned with what is happening at a given moment but with what will happen at the next. This leads to rapid succession of the sequential movements one after the other to produce a smooth, joined complex movement.

4. **Servocomparator (force regulator):**

- The spinocerebellum receives input signals from:

a) The premotor cortex and inferior olive, before the start of movement. An "efference copy" of the plan of movement informs the cerebellum of the "**intention**" to make a movement.

b) The spinocerebellar tracts, during the movement. An "afference copy" of signals informing the cerebellum about the "**performance**" of the movement.

- The spinocerebellum then compares the afference with the efference copy. If there is any error in the performance, the cerebellum sends **correction** to adjust the movement to match the intention.

5. **Damping of movement:**

- The spinocerebellum acts as a "**brake**" which terminates the movements exactly at the right time when the desired movement is completed. The absence of this function leads to "**intention tremor" or "kinetic tremor**".

6. **Predictive function:**

- As the cerebrocerebellum receives sensory input from all sensory organs, so the cerebellum uses this information to compute the rate and direction of movement to reach a certain target at a certain time; e.g. the thrust of a goal-keeper to catch a kicked football, or the thrust of a tennis player to meet a coming ball with his racket.

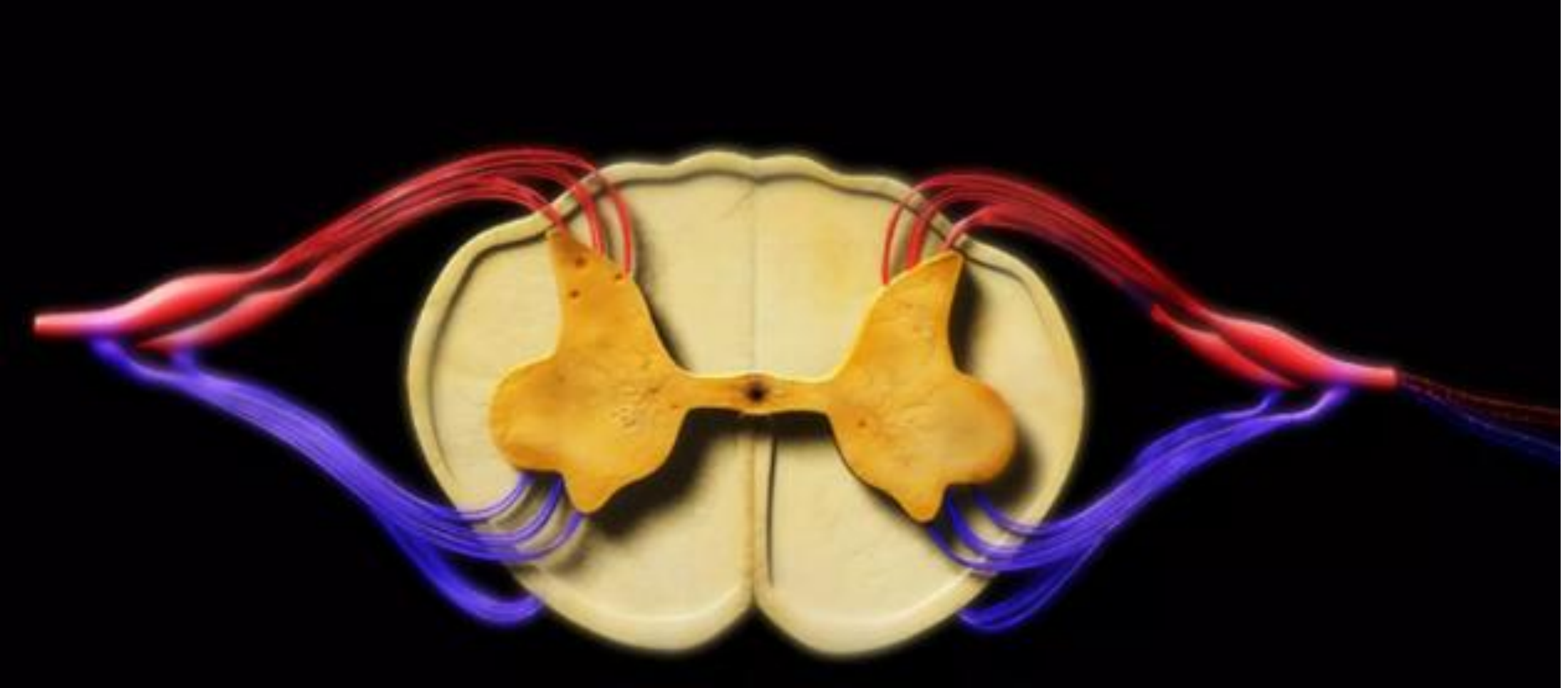
7. **Initiation and termination of ballistic movement:**

- A ballistic movement is a short movement whose direction and speed is determined by the initial thrust (finger typing). The role of the cerebellum here is to potentiate the starting of the movement to allow a rapid onset and to terminate the movement abruptly without overshooting.

The Cerebellar Syndrome

- The cerebellar syndrome results from lesions that involve the cerebellar nuclei.
- Each cerebellar hemisphere control the contralateral cortical motor area, so the cerebellum exerts its effect on the same side of the body. This syndrome include three main features:
  asthenia, atonia and ataxia (AAA)

1. **Asthenia (muscle weakness):** difficulty in initiating and maintaining muscle contraction because the function of potentiating motor signals is lost.
2. **Atonia (or hypotonia):** lack of muscle tone. This occurs due to loss of the facilitatory effect.
3. **Ataxia:** incoordination of voluntary movements.



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